SYED MUHAMMAD SHARIQ

AI & Computer Vision Systems Engineer

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TECHNICAL SKILLS

Languages & Core: Python (Advanced), C++, Java, SQL, Bash, Linux / Unix

Computer Vision & ML: PyTorch, YOLO (v8/v11), OpenCV, EasyOCR, TensorFlow, MediaPipe, Edge Device Optimization (TensorRT/ONNX), Real-Time Video Streaming (RTSP)

AI Systems & Backend: FastAPI, WebSockets, Google Vertex AI (Gemini 2.5 Flash), AssemblyAI Voice API, Pydantic v2, Asyncio, Edge-TTS, REST APIs

Data & DevOps: Dataset Curation & Annotation (Roboflow, CVAT), Docker, Docker Compose, Git, CI/CD (GitHub Actions), Google Cloud Run, SQLite

WORK EXPERIENCE

LA Consulting Corp — AI Engineer (Computer Vision & Edge Optimization) Nov 2025 – Present | Karachi, Pakistan

- Architected and benchmarked lightweight **YOLOv8 models optimized for edge hardware** and embedded systems, sustaining **30+ FPS** across live video feeds.
- Automated multi-camera RTSP ingestion and preprocessing pipelines across **10,000+ custom annotated frames**, slashing model training preprocessing turnaround by **35%**.
- Collaborated with senior engineers to implement zero-latency tracking and recognition features, integrating automated stress-testing and CI validation workflows.

State Bank of Pakistan (BSC) — Technology & Operations Internee June 2026 – August 2026 | Karachi, Pakistan

- Audited public procurement and enterprise compliance documentation in adherence with PPRA (Public Procurement Regulatory Authority) standards.
- Coordinated cross-departmental technology event operations, resource allocation, and internal system documentation.

Zam Zam Batteries — Operations & Inventory Systems Assistant Feb 2023 – July 2025 | Karachi, Pakistan

- Digitized inventory logs and client sales records, transitioning manual ledgers to structured data systems to eliminate stock discrepancies and improve order fulfillment accuracy.

FEATURED PROJECTS

Intelligent Traffic Violation Detection System FINAL YEAR PROJECT / CAPSTONE github.com/Shariq-duet/...

Tech: *Python, PyTorch, YOLOv8, OpenCV, EasyOCR, Roboflow, Pandas*

- **Dataset Author:** Curated, labeled, and published **2 open-source benchmark datasets on Roboflow & Kaggle** (Helmet compliance & Driver distraction) to solve critical zero-shot training gaps in regional traffic scenarios.
- Engineered an end-to-end multi-model enforcement pipeline tracking vehicles in custom ROIs to detect **5 violations**: Helmet Compliance, Seatbelt Usage, Phone Distraction, Red-Light Jumping, and Vehicle Closeness/Tailgating.
- Implemented heuristic crop selection (`best_SS.py`) scoring sharpness/contrast before feeding plates into **EasyOCR** for automated license reading, achieving **91.4% mAP@50** with real-time multi-lane CCTV inference.

CallBridge — Autonomous Voice Negotiation Agent with HITL Safeguards github.com/Shariq-duet/CallBridge

Tech: *Python 3.12, FastAPI, WebSockets, AssemblyAI Voice Agent, Microsoft Edge-TTS, Pydantic v2, Docker*

- Architected an outbound voice agent negotiating phone calls with strict **Human-in-the-Loop (HITL) approval gates**, preventing unauthorized price commitments and personal data disclosures.
- Engineered a hybrid speech pipeline pairing AssemblyAI's full-duplex Voice Agent API (live turn detection, holding phrases) with free Microsoft edge-tts neural voice generation (322 voices) and in-memory audio caching.
- Implemented thread-safe session state machines with optimistic concurrency conflict detection (HTTP 409) and containerized deployment with non-root Docker and automated CI checks.

Game Debugger — Agentic Multi-Source Bug Triage Pipeline github.com/Shariq-duet/game-debugger-agent

Tech: *Google Vertex AI, Gemini 2.5 Flash, Google Cloud Run, Pydantic v2, Discord API, Python*

- Deployed a 3-phase autonomous agent pipeline on **Google Cloud Run** ingesting noisy multi-source community bug reports across Discord, server logs, and CSV uploads.
- Engineered a cross-source contradiction-detection algorithm that scores reporter credibility and reconciles conflicting reports prior to generating reproducible Jira tickets and automated code patches.

ChangeProof — AST-Based Architectural Invariant Verification Engine github.com/Shariq-duet/ChangeProof

Tech: *Python (ast), Static Code Analysis, GitHub Actions, CI/CD*

- Built an automated linter using Python's native Abstract Syntax Tree (`ast`) module to enforce Architectural Decision Record (ADR) invariants during pull request CI builds, preventing architectural drift and boundary leaks before merge.

EDUCATION

Dawood University of Engineering & Technology (DUET) — Bachelor of Science in Computer Science (BSCS) 2023 – Expected May 2027 | Karachi, Pakistan